

Question	Marking details	Marks available																				
		AO1	AO2	AO3	Total	Maths	Prac															
7	Other nucleic acids involved in protein synthesis:	6		3	9																	
	<table><tr><th>Nucleic acid</th><th>role</th><th>Where it functions</th></tr><tr><td>DNA</td><td>provides the {genetic code/ template} for production of {all RNAs/ mRNA/ provides the genetic code for sequence of amino acids}</td><td>nucleus</td></tr><tr><td>rRNA</td><td>To form ribosomes/ the site of translation</td><td>cytoplasm/ rough ER</td></tr><tr><td>mRNA</td><td>Produced in transcription and used in translation/ transfers genetic code (from nucleus to ribosomes) / owtte</td><td>cytoplasm/ ribosomes</td></tr><tr><td>tRNA</td><td>delivery of amino acids to the site of protein synthesis</td><td>cytoplasm</td></tr></table>							Nucleic acid	role	Where it functions	DNA	provides the {genetic code/ template} for production of {all RNAs/ mRNA/ provides the genetic code for sequence of amino acids}	nucleus	rRNA	To form ribosomes/ the site of translation	cytoplasm/ rough ER	mRNA	Produced in transcription and used in translation/ transfers genetic code (from nucleus to ribosomes) / owtte	cytoplasm/ ribosomes	tRNA	delivery of amino acids to the site of protein synthesis	cytoplasm
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Section C: How tRNA activation works: • ATP provides energy for the reaction • Two phosphates are released • Amino acid reacts with ATP to form activated amino acid. Amino acid is transferred to tRNA forming activated {tRNA/ amino acid}																						